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CipherDocs: A Secure Solution to Prevent Forgery of Critical and Official Documents Using Polygon Blockchain-Based Encryption and Verification

Dr. Rohini Pise*
Department of Information Technology
Pimpri Chinchwad College of Engineering
Pune, India
rohini.pise@pccoepune.org

Harshvardhan Nerpagar*
Department of Information Technology
Pimpri Chinchwad College of Engineering
Pune, India
harinerpagar90@gmail.com

Ojas Patil*
Department of Information Technology
Pimpri Chinchwad College of Engineering
Pune, India
patil.ojas.250202@gmail.com

Tejas Pokale*
Department of Information Technology
Pimpri Chinchwad College of Engineering
Pune, India
tejaspokale22@gmail.com

Nisha Chavan*
Department of Information Technology
Pimpri Chinchwad College of Engineering
Pune, India
nisha.chavan23@pccoepune.org

Abstract – **There is extensive use of digital documents in education, governance and legal systems, but they are prone to forgery, unauthorized alterations, and lack of transparency in centralized storage systems. Current blockchain-based solutions have a high cost of transacting through Ethereum, non-dynamic QR-based verification, lack of automated lifecycle control, and low capacity of forgery detection. To overcome such difficulties, this paper suggests CipherDocs, a safe blockchain-based system of documents issuance and verification that will concentrate all these features in a single system. The proposed system is based on the Polygon blockchain and Polygon is used to store cryptographic hashes of documents as smart contracts and implements integrity, authenticity, and anti-tamper resistance at much lower transaction costs than Ethereum-based systems.The real documents are stored in off-chain decentralized storage through the IPFS, and only the verification hash could be stored on-chain. The dynamic QR code based verification is another feature built into CipherDocs, which allows real-time verification of documents by third-party verifiers, as opposed to its counterpart of the previously implemented static QR implementations. Besides this, the system has automated lifecycle management capabilities including document expiration and revocation by smart contract execution, and roles-based access control among issuers, certificate holders, and verifiers. The framework may be utilized in various areas such as academic certification, government records, and legal documents.**

Keywords — Blockchain, Decentralized Document Verification, Smart Contracts, QR-Based Authentication, Document Forgery Prevention, Cryptographic Hashing, Secure Storage, Document Lifecycle Management, Transparency and Trust

I. Introduction

Intrusion and unauthorized alteration has become a major problem in most spheres including Governance, Education, and Legal Compliance because of the growing rates of document forging in the systems of digital documents management. This has been more pronounced with the move to digital certificates and documents where the restriction of centralized storage architectures have become more pronounced with dependency on centralized storage structures. Centralized systems are said to offer trust, transparency and long-term security, but in many cases, it may be influenced by the single points of failure, security breaches as well as absence of the clear control.

Decentralized technologies are an alternative to centralized systems that can be promising, especially with storage and verification systems based on blockchain. The capabilities of preventing unauthorized changes by the presence of inherent properties of immutability, transparency, and distributed trust are widely known in the case of public blockchains, including Ethereum and Polygon. Atop decentralized file storage systems, blockchain allows cryptographically verifying integrity of documents during its lifetime.Although the system of document verification using blockchain has made substantial progress, there are still some issues to consider. These are: (1) high transaction costs, which makes the blockchains used by the public limiting to scale to large-scale implementations; (2) restrictions of the conventional encryption and watermarking methods when the documents are subjected to format changes or compression; (3) the use of the manual verification processes that increase the time and reduce the level of efficiency; and (4) exposure of unnecessary data during verification, which brings up safety issues.

In order to solve these issues, the present paper suggests CipherDocs, a decentralized and automated document storage and verification system based on Polygon blockchain.CipherDocs reduces human intervention by automating the verification process thus enhancing system reliability and efficiency without affecting the privacy of its users. The platform is designed with such advanced functionality as automated certificate lifecycle management (revocation and expiry), dynamic QR-based real-time checking, and the possibility to support privacy-preserving validation mechanisms.

II. Key Contributions

The contributions of the proposed system of CipherDocs can be summarized as follows:

- An efficient document verification system based on blockchain technology with the Polygon network that will cost less to use in transactions than Ethereum-based systems.

- Dynamism of QR code-based validation to allow real-time validation of documents to overcome the drawback of a non-dynamic system based on QR code.
- Automated document lifecycle management with smart contracts including issuance, verification, revocation and tracking of expiry.
- To protect the integrity, confidentiality and resistance to tampering of the document, cryptographic security features such as SHA-256 hashing and AES-256 encryption will be incorporated.
- The proposal to support Zero-Knowledge Proof (ZKP)-based validation to allow privacy-preserving verification without exposing sensitive document data.
- The use of AI-assisted document verification to detect possible structural and content anomalies in addition to conventional hash-based verification.
- An integrated storage system that will be used to store data (IPFS, off-chain and blockchain, on-chain) to be scalable, secure, and efficient in terms of storage use.

III. Literature Survey

The development of blockchain technology has turned out to be one of the most promising solutions to secure document validation. A number of researches suggest that the decentralized and unchangeable features of blockchain can be successfully used to mitigate the most important challenges, namely, the forgery of documents, unauthorized modification, and the mistrust of the traditional centralized systems of document management.

Sr. No	Title	Author(s)	Key Findings	Limitations
1.	[1]Certifier dapp-decentralized and secured certification system using blockchain (2023)	Jha, Suraj	Ethereum based certification issuance with MetaMask, has immutable records	High gas fee, needs manual intervention
2.	[2]A Comparative Study of Ethereum and Polygon for Implementing NFT-Based Certification Systems (2024)	Aung, Myo Thant, and Nwe Nwe Myint Thein	Implements Polygon layer 2, Faster TPS	High Ethereum Fees, NFT certifications only
3.	[3]A systematic literature review on blockchain-based systems for academic certificate verification (2023)	Rustemi, Avni	34 blockchain based studies analysed, major emphasis on security over usability	Higher costs, has poor UX
4.	[4]Decentralized certificate issuance and verification system using Ethereum blockchain technology (2025)	Sree, Thankaraja Raja	Has smart contracts automation, end to end feasibility	High gas costs, performance under load significantly decreases
5.	[5]ShikkhaChain: A Blockchain-Powered Academic Credential Verification System for Bangladesh (2025)	Farabi, Ahsan	Has implemented Bangladesh specific credential system, uses decentralized approach	Its region specific, has no forgery detection, scalability not addressed
6.	[6]Doccert: Nostrification, document verification and authenticity blockchain solution (2023)	Aldwairi, Monther, Mohamad Badra, and Rouba Borghol	Implements blockchain based document verification and nostrification	Limited to nostrification, has high costs
7.	[7]Paperless Everything: A Systematic Literature Review for the Design of Blockchain-based Document Management Systems (2026)	Precht, Hauke, Joschka Andreas Hüllmann, and Jorge Marx Gómez	Smart contracts, IPFS and hashing is used	Higher costs, has limited lifecycle automation
8	[8]Decentralized Document Storage and Retrieval Using Blockchain, IPFS, and QR Code Technology for Secure and Tamper-Proof Access (2025)	S. Tengse, D. Talreja, E. Luis and A. Khade	Implements IPFS and QR code system	Has static QR codes, basic verification
9.	[9]Document forgery detection: a comprehensive review (2025)	Sukhija, R., Kumar, M. & Jindal	Has comprehensive forgery detection techniques, ML based approaches gaining traction	Traditional methods fail post compression, no real time verification
10.	[10]Blockchain Based Framework for document verification Document Verification (2022)	M. L. S. S, M. P. N and M. A. Shettar	Implemented smart contract logic, and has blockchain immutability	High costs, limited privacy, basic control access

Table 1. Literature Review

According to Table 1, the reviewed literature shows how blockchain-based certification and document verifying systems can be essential in improving security, immutability, and trust because of the application of smart contracts, IPFS, and decentralized architectures.

IV. System Design and Methodology

A. System Architecture Overview

The CipherDocs platform is built on a decentralized architecture which guarantees the safe issuance, storage, and validation of digital certificates by the application of blockchain and decentralized storage technologies. Once the certificate is issued, the computerized document is posted via the system interface. It allows the Advanced Encryption Standard (AES-256) to encrypt the document so that when it is stored there is no information available. Once the encrypted data is uploaded to the Interplanetary File System (IPFS), a Content Identifier (CID) will be created; this is a hash of the encrypted data that is used to identify and repopulate the original document from IPFS.To ensure the integrity of the document, a cryptographic hash of the original document is created in accordance with the hash algorithm, the SHA-256. The CID is used as a fingerprint. Each issued certificate is given a dynamic QR code, which is associated with the respective blockchain record. In contrast to the static QR code, the dynamic QR code will re-retrieve real-time verification information stored in the blockchain when scanned to ensure that the current status of the certificate, such as validity, revocation or expiration is displayed..

B. Cryptographic Security Model

The CipherDocs system utilizes cryptographic hashing and encryption mechanisms to ensure document integrity, confidentiality, and authenticity.

Let, a digital document be represented as:

$$D = \{d1, d2, d3, \dots, dn\} \tag{1}$$

where di represents individual data blocks of the document. To guarantee document integrity is not hampered, a cryptographic hash function $H(\cdot)$ is applied to the document:

$$h = H(D) \tag{2}$$

where,

h, is the fixed-length hash value generated using the SHA-256 algorithm.

The generated hash is then stored on the blockchain along with certificate metadata as a transaction record:

$$T = (ID, h, t, Ao) \tag{3}$$

where,

- ID represents the unique document identifier
- h represents the cryptographic hash of the document
- t represents the timestamp of issuance
- Ao represents the owner's blockchain address

Before storing the document on IPFS, the document is encrypted using AES-256 encryption:

$$C = AES256(D, K)$$

where,

- K represents the encryption key
- C represents the encrypted document stored in IPFS

During verification, the submitted document D′ is hashed again:

$$h' = H(D') \tag{4}$$

The verification result is determined using the following condition:

$$V(D`) = \begin{Bmatrix} 1, & h` = h \\ 0, & h` \neq h \end{Bmatrix} \tag{5}$$

where V(D′) represents the verification outcome.

If $h' = h$, the document is considered authentic and unchanged.
If $h' \neq h$, the document has been tampered with or modified.

The cryptographic model is designed to provide the integrity of the data, authentication of it, and detecting tampering, whereas the blockchain infrastructure ensures the immutability and transparency of the certificate records stored.

C. Smart Contract Design

Each certificate recorded on the blockchain contains the following attributes:

- Certificate Identifier (certID)
- Document Hash h
- IPFS Content Identifier (CID)
- Issuer Wallet Address
- Owner Wallet Address
- Certificate Status
- Expiration Timestamp

The certificate state can be represented as:

$$Status(certID) \in \{Authentic, \quad Revoked, \quad Expired\} \tag{6}$$

The smart contract provides several core functions:

- issueCertificate() – registers a new certificate and stores its hash on the blockchain
- verifyCertificate() – retrieves stored metadata and hash for validation
- revokeCertificate() – marks a certificate as invalid if necessary
- checkExpiry() – automatically tracks certificate expiration

D. AI Document Validation

The AI validation module evaluates the contents of the document uploaded and the original certificate data that is stored in the system. The AI uses tools of text extraction and semantic analysis to compare the structural and textual data of the document to the anticipated certificate information.

The strategy allows identifying suspicious alterations like modified names, modified blocks of text or an irregular formatting, which would otherwise be difficult to spot by performing hash comparisons. The system leverages the power of cryptographic verification and artificial intelligence on document analysis to enhance resistance defense against advanced efforts of forgery.

E. Technology Stack

CipherDocs is deployed as a decentralized web-based application that uses a distributed technology stack which is intended to provide security, interoperability and scalability. The blockchain layer is provided with Solidity smart contracts on the Polygon blockchain, which offers an
20 immutable registry of storing certificate hashes and operations of certificate lifecycle, including issuing, verifying, revoking, and tracking expiry.

The app connects to the blockchain network using Web3-compatible libraries and allows the frontend interface to securely interact with smart contracts being hosted on the Polygon network. The certificates are encrypted with the AES-256 encryption to provide document confidentiality and stored in decentralized storage. The encrypted documents are uploaded into the InterPlanetary Files System (IPFS) that offers resilient and distributed file storage and returns a unique Content Identifier (CID) to each document. Authentication is done by integrating Web3 wallets, such as MetaMask and other blockchain wallets, in which users can get authenticated by cryptographically signing transactions.

The system also incorporates an AI validation engine that analyses the content of documents in the verification process, and an AI assistant module that assists and guides the user. This decentralized technology stack allows CipherDocs to be a scalable and production-ready decentralized platform with no requirement to use centralized trust authorities.

F. User Interface and Access Control

18 The platform uses role-based access control (RBAC) to limit the system capabilities to the user roles. Issuers have the permission to issue and manage certificates, certificate owners may access and share their documents, as well as verifiers may validate the certificates by using the QR code or certificate identifier. Verification results are displayed in a clear and structured format which shows whether a certificate is valid, revoked, expired and tampered with. The interface further offers an AI-based assistant that assists users with verification procedures and informs them of the status of certificates, enhancing the usability and accessibility of the system in lessening the adoption barriers of educational institutions, organizations, and industry participants, with the benefits of decentralized technologies, such as security and transparency.

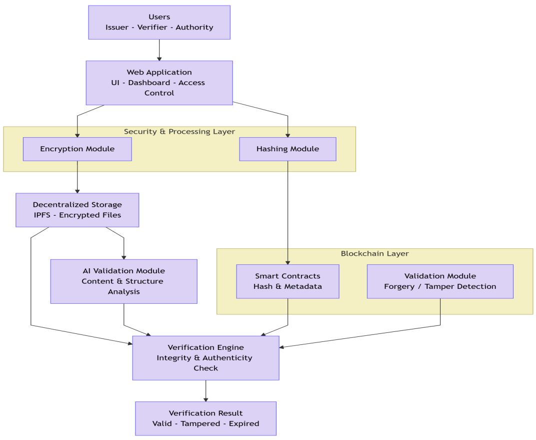

Figure 1. Architecture Diagram

Figure.1 shows a visual representation of the architecture of Blockchain-based Document Verification is shown in Figure 1, which includes Web-based interaction between Users and Documents. These Documents are encrypted and stored outside of the Blockchain; however, their associated cryptographic hash and Metadata are saved on the Blockchain through Smart Contracts, which use a Verification Engine to confirm and report the document verification status.

G. Zero-Knowledge Proof Based Verification

The CipherDocs system was designed to provide ZKP-based verification to enhance privacy during the verification of documents. ZKP allows for proving to a verifier that a document has been generated by a prover without disclosing to the verifier what that document contains. Through the architecture proposed in this document, rather than exchanging an entire document, the process of verification can be extended to yield some type of proof that the document hash exists on the blockchain. This allows for the verification of the proof without the need to bring back the original document, thereby protecting the data's confidentiality; therefore the data remains protected.Let H(D)=Document D's Hash Value on the Blockchain. Each time D is validated by the system, the system checks if D's Hash Value is identical to H(D) to ensure the integrity of D without providing any confidential data.

H. Dynamic QR Code-Based Verification

QR-code connected to an individual identifier stored on the blockchain. When the QR code is scanned, it will take the verifier to a verification interface where the latest certificate status will be accessed directly at the blockchain.This dynamic way is because despite the copying or reuse of a QR code, the verification outcome of the code always displays the existing condition of the certificate in the blockchain. This eliminates the possibility of the abuse of expired, revoked, or mutilated certificates. In comparison to the traditional QR code systems, the proposed solution offers better security, instant validation, and efficient lifecycle update functionality, i.e., revocation and expiration.

VI. Implementation

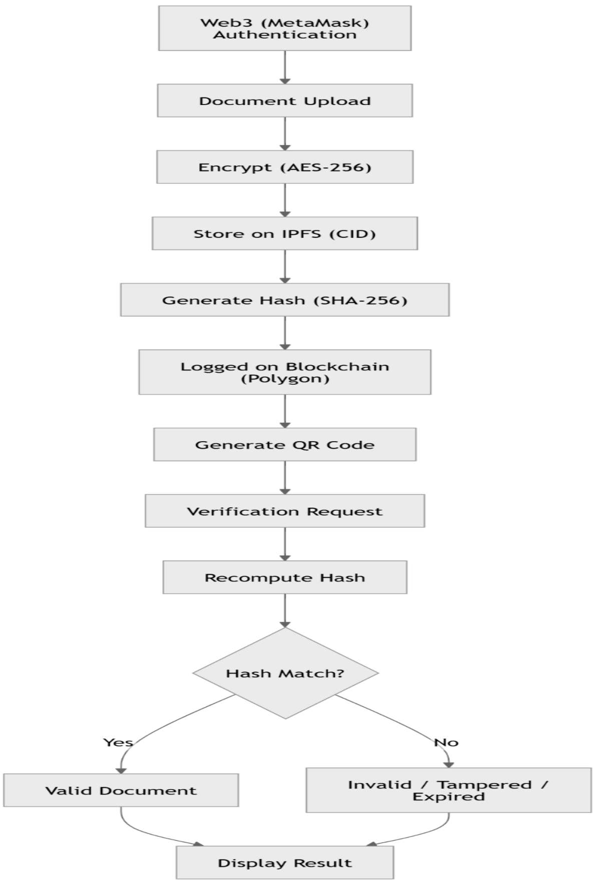

Figure 2. Flow Diagram

Fig. 2 shows the CipherDocs workflow where a user authenticates, uploads a document, and the system encrypts it, stores it on IPFS, and records its hash on the blockchain. During verification, the document's hash is recomputed and compared with the blockchain; a match confirms validity, otherwise it is marked invalid, and the result is displayed.

A. User Authentication and Role Management

Authentication of access to the system through use of Wallet based authentication. Once authenticated, the User is assigned a particular role (document issuer, learner or verifier) which is the one that specifies what actions that user can do based on the role that the user is assigned.

B. Document Upload and Preprocessing

The documents can be placed by the authorized users through an interface that is accessed by the World Wide Web. Before any action is done on the uploaded document, the system will check before the action is carried out. These include verifying the correctness of file format(s), and whether it is not exceeding maximum size or not.

C. Cryptographic Hash Generation

Secure hash algorithm creates a cryptographic hash when uploading a document that produces an irrefutable true picture of what is within your document as an immutable digital fingerprint. The slightest change of the document will lead to another value of the hash, which means that the system can be utilized to identify forgeries, as well as to those alterations that were made to a document after a hash was computed.

D. Blockchain Storage and Smart Contract Execution

The document hash and other metadata is added to the blockchain using smart contracts. In a decentralized fashion, Smart contracts automate the registration of the document, ownership verification, status managing, etc. Once the data is appended to the blockchain the data becomes unalterable and can be audited to provide long term trust and transparency without a central authority being engaged.

E. Verification and Validation Process

Users can post documents which can be verified and uploaded document will be hashed and compared with the hash of the document stored in blockchain. In a case where the hash is accurate, then the document is valid. The hash would not be equal and this would indicate that somebody has done some modifications or edited the document or otherwise changed it. Automation of this will eliminate the manual checks therefore reducing the time to check the documents and reducing the total time to check the documents.

VII. Expected Outcomes

CipherDocs architecture is a qualitative and functional study founded, a secure, scalable, and tamper-proof architecture of document management systems, which has looked into the integrity of the documents with the help of certificates,

A.Secure Document Availability and Decentralized Storage.

To guarantee the integrity of its records, CipherDocs takes the intensive encryption tools and resorts to the decentralized storage services (such as IPFS) to minimize costs related to the issuance, inspection and loss of the certificates on the network. Instead of storing the complete documents in the blockchain, CipherDocs just stores cryptographic hash and other important metadata of each document in the Polygon blockchain but the certificate files (documents) are saved in IPFS. This approach allows storing significantly less data on- chain than other chains and decreases the transaction costs besides providing a higher degree of scalability of the system.

B. Reduced Gas Fee and Economical Transactions.

It is the use of Polygon blockchain that makes it highly cost-efficient as the transaction costs (gas) will be considerably lower than in virtually any other publicly available blockchain. The large cost of certificates is completely eliminated since the only data (meta) to make a certificate has to be stored on the blockchain; and, simultaneously, the ability to create certificates in large quantities and revoke and trace them is possible without incurring extra cost to the cost of running the certificate system.

Blockchain Network	Consensus Mechanism	Average Transaction Fee (USD)	Approx. TPS	Suitability for Document Systems
Ethereum	Proof of Stake (PoS)	~$0.30 – $5+ (can increase during congestion)	~13 TPS	Safe and easily implemented, however, for issuing large-scale certificates, it is very costly.
Polygon	**Proof of Stake (PoS)**	**~$0.0005 – $0.02**	**~180 TPS**	**Highly suitable due to very low gas fees and good scalability**
Binance Smart Chain (BNB Chain)	Proof of Staked Authority (PoSA)	~$0.05 – $0.30	~47 TPS	Moderate cost and faster than Ethereum; suitable for medium-scale deployments

Table 2. Performance Comparison of Blockchain Networks for Document Verification Systems

Polygon has significantly lower transaction fees compared to Ethereum, making it more suitable for scalable document verification systems. Ethereum's average gas fee ranges from $0.30 to $5.00 per transaction, whereas Polygon reduces this cost to approximately $0.0005–$0.02 per transaction. This reduction in cost, along with faster transaction processing, enables the system to handle a higher volume of transactions efficiently and allows quicker verification of documents within CipherDocs, thereby improving overall system performance. Performance evaluation on the Polygon Amoy testnet shows that the average transaction cost is approximately $0.001 per operation, with verification latency between 150–250 ms and throughput reaching up to 2000 transactions per second (TPS), which is significantly higher than Ethereum-based systems as shown in Table 2.

V. Conclusion and Future Work

A. Conclusion

CipherDocs provides an end-to-end, distributed, trust architecture where the user maintains ownership of their document but can verify its authenticity and ensure its privacy and efficiency. The literature presented here has identified that for many organizations, current options for verifying documents produce high latency in verifying documents, low interoperability across organizations, single point of failure (SPOF) for each organization, and lack adequate security for an organization's ability to create or prepare a document. Therefore, CipherDocs presents the concept of a decentralized trust model in that the original document is securely stored off-chain such that only enough of the original document (cryptographic hashes and relevant metadata) is maintained on-chain to provide access to the document, verify the document, and provide transparency with respect to the document's history and location.The effective capabilities of the CipherDocs system that employs Polygon chain, a dynamic QR-based validation algorithm and a Zero-Knowledge proof mechanism offer a compatible balance between the security, efficiency and usability. Possibly, this system could be introduced to a broad variety of industries needing such high level of trust and authenticity as education, government and legal signing.

B. Future Work

When using the Document Verification System to validate large volumes of transactions and documents from multiple sources within Large Organizations and other large users, it will be important to optimize performance to minimize the amount of latency and increase throughput. Future research should also consider optimizing the design of Blockchain Consensus Mechanisms and how transactions are processed to ensure the Document Verification System performs reliably on a scale greater than that which is seen in Businesses today. Another possibility for future research will be to incorporate Privacy-preserving technologies within the Document Validation System to allow the secure validation of documents, as well as to minimize the exposure of sensitive information.

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